

Guideline

Vancomycin - Intravenous

Reference #: MMG008

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Medical, Nursing, Pharmacy

HIGH RISK MEDICATION

Vancomycin is an irritant drug. Extravasation may cause tissue necrosis.

Vancomycin can cause severe infusion reactions including profound hypotension and severe infusion related reactions.

Monitoring requirements exist for this medicine to minimise toxicity and optimise therapeutic efficacy

Introduction

Vancomycin is a widely used intravenous antibiotic used for the treatment of Gram-positive pathogenic bacteria.

It works by inhibiting cell wall synthesis in susceptible bacteria to inflict a bactericidal effect on the affected cell via osmotic shock and cell lysis.

Despite its efficacy, vancomycin can cause some potentially toxic adverse effects, especially when administered in parenteral form. Some potential effects include nephrotoxicity, hearing impairment as well as infusion-related reactions.

It is recommended that close therapeutic drug monitoring is performed to provide guidance with dosing and overall use of this medication.

Definitions

Anephric	Defined as a medical condition where there is the absence of functioning kidneys to adequately to filter or remove waste products from the blood
CVVHDF	Continuous venovenous haemodiafiltration – a form of continuous renal replacement or dialysis that combines the principles of haemodialysis and haemofiltration to clear blood solutes by use of diffusion and convection processes
Loading Dose	The use of a single large initial dose or series of larger doses administered at the onset of therapy, with the intention of reaching the target therapeutic concentration in the body rapidly
Trough level	A trough drug level, or concentration refers to a level taken which represents the concentration of the drug in the blood immediately before the next dose is administered.

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Presentation (1)

Glass vial containing vancomycin as a lyophilised powder (white to light brown powder or plug) for reconstitution prior to administration.

Once reconstituted using water it will form a clear solution with pH between 2.5 to 4.5.

Action & Clinical Effects (1, 2, 3))

Bactericidal glycopeptide antibiotic that inhibits bacterial cell wall synthesis by preventing formation of peptidoglycan polymers². Essentially it interferes with the extension of the growing peptidoglycan chain

resulting in the formation of an unstable bacterial cell wall which will be vulnerable to osmotic pressures and undergo cell lysis/death.

Indications (4)

- Treatment of serious gram-positive infections (eg MRSA, Coagulase-negative Staphylococci infections)

There are many indications for the use of vancomycin. An extensive list of approved indications is available from Formulary One ([WA Statewide Medicines Formulary \(health.wa.gov.au\)](http://health.wa.gov.au))

Additionally, eTG : Antibiotic is a useful reference for guidance on appropriate indications for intravenous vancomycin.

Commencement of vancomycin should be on the advice of an ID Physician or Clinical Microbiologist.

Precautions

- Vancomycin is a restricted antimicrobial and requires approval from the Infectious Diseases teams (ID phone approval, ID formal consult, ID MDT meetings, HAPI advice) or Clinical Microbiology for use in patients at SCGOPHCY.
- Careful therapeutic drug monitoring should accompany vancomycin use to minimise the potential toxicities associated with the use of this medication and to optimise therapeutic efficacy.
- Extra caution should be applied when prescribing vancomycin for patients with renal dysfunction, presence of concomitant nephrotoxic agents (eg aminoglycosides, colistin, cisplatin, piperacillin-tazobactam), the elderly or patients with pre-existing hearing impairment.
- Concurrent use of combination therapy with **vancomycin and piperacillin-tazobactam** has been associated with a **3 to 4-fold increase** in the risk for **nephrotoxicity**. Alternative options are available to use, however if no suitable alternatives available then the duration of this combination therapy should be kept to the minimum duration possible.
- Avoid rapid intravenous administration (eg bolus injection of vancomycin over several minutes only) as rapid administration increases the risk for *vancomycin infusion reaction (VIR)* – formerly known as ‘red-man syndrome’ (for more information see ‘Adverse Effects’ section) leading to exaggerated hypotension, flushing, erythema, urticaria, pruritus and rarely cardiac arrest.

VANCOMYCIN QUICK REFERENCE GUIDE

HIGH RISK MEDICATION

This is a high-risk medication. Strict monitoring and administration requirements apply to the use of the medication

DRUG	PRESENTATION	DOSAGE	DILUTION VOLUME	ADMINISTRATION RATE	THERAPEUTIC DRUG MONITORING
Vancomycin	Glass vial containing lyophilised powder (white to light brown)	LOADING DOSE If deemed necessary 25 to 30mg/kg (ABW) as single dose Max 3000mg	The final concentration of the solution should be 5 to 10mg/mL Final concentration should NOT exceed 10mg/mL, even in fluid restricted individuals	PERIPHERAL ACCESS Recommend 10mg/min May be increased up to 16.66mg/min but only if necessary and patient tolerating	Collect trough levels after steady state concentration has been reached (eg after 4-5 doses have been given) Target level 15-20mg/L (up to 25mg/L) Collect sample using DARK GREEN or RED top  Following sampling, do NOT wait for the result to return before giving next dose unless specifically advised
	PREPARATION Dissolve each 1g vial using ~20mL WFI. This must be further diluted See administration guide below	MAINTENANCE Dependent on renal function See dosing table below		CENTRAL ACCESS Recommend 10mg/min with option to increase to 16.66mg/min See administration guide below	

DOSING GUIDE (SUMMARISED)

CrCl* (mL/minute)	Suggested dose# ^	Timing for first trough* concentration level collection
Greater than 60mL / min	15 to 20mg/kg 12-hourly	Before the fifth dose
Between 40 to 60 mL /min	15mg/kg 12-hourly	Before the fifth dose
Between 20 to 39mL /min	15 to 20mg/kg daily	Before the fifth dose
Less than 20mL / min	15mg/kg every 48 to 72 hours	Before the second dose is due NB. This level is used for guiding when to re-dose
* Creatinine clearance is generally based on estimates using the Cockcroft-Gault formula		
# Dosages suggested in this table are serve as a guide only for the initial doses prior to trough concentrations becoming available. Many other influencing factors are not taken into consideration in the above guide. Use ABW for dosage calculations		

ADMINISTRATION GUIDE (SUMMARISED)

Type of parenteral access available	Dose	Minimum Volume	Recommended Rate
Peripheral intravenous cannula (PIVC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes
	1.25g to 1.5g	≥ 250mL	Infuse over at least 90 minutes
	2g	500mL	Infuse over at least 120 minutes
Central venous access device (eg CVC, PICC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes
	1.25g to 1.5g	≥ 250mL	Infuse over at least 90 minutes
	2g	≥ 250mL	Infuse over at least 120 minutes
AVOID SUBCUTANEOUS, INTRAMUSCULAR OR BOLUS IV INJECTION			

DOSE ADJUSTMENT GUIDE (SUMMARISED)

Vancomycin trough level (mg/L)	Recommended dose adjustment for next dose (to achieve target vancomycin level 15 to 25mg/L)
>40	Withhold the next dose and repeat level before timing of next scheduled dose Investigate for contributing factors/potential causes for the high vancomycin level Refer to the guideline section 'vancomycin dose adjustment guide' for advice
>30	Withhold the next dose and repeat level before timing of next scheduled dose Refer to the guideline section 'vancomycin dose adjustment guide' for advice
25 to 30	Review renal function / signs of evolving renal dysfunction If no evidence of developing renal dysfunction then continue with next scheduled dose but reduce by 1/3 of the original dose (eg usually 250 to 500mg)
15 to 24 (target range)	No change required. Continue current dose.
10 to 14	If target level is 20-25mg/L, consider increasing the total daily dose by 500mg
< 10	Increase the total daily vancomycin dose by 750 to 1000mg

If your patient experiences an **infusion-related reaction**, reduce the rate of the infusion and refer to the section on 'Management of Vancomycin-infusion reactions'

If your patient experiences a **severe reaction such as anaphylaxis**, STOP the infusion immediately and seek medical help

VANCOMYCIN QUICK REFERENCE GUIDE

HIGH RISK MEDICATION

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DOSING IN ADVANCED RENAL IMPAIRMENT (DIALYSIS)

SEVERE ACUTE RENAL IMPAIRMENT

Severe renal impairment but not requiring dialysis

1. Consider the use of a loading dose (25 to 30mg/kg) when starting vancomycin therapy
2. A subsequent dose of 15mg/kg to be given only when the vancomycin level falls to $\leq 15\text{mg/L}$
3. Perform repeat plasma vancomycin levels every 24 to 48 hours to determine timing for redosing

Based on clinical experience the doses will be approximately every 48 to 72 hours

HAEMODIALYSIS (HIGH FLUX)

Patients receiving high flux haemodialysis

1. Initial loading dose between 15 to 20mg/kg upon starting therapy
2. Vancomycin level in 48 to 72 hours (timing of level to align with their next HD session). Level to be taken **before** their dialysis on that day.
3. Depending on the vancomycin result, a vancomycin 'top up' dose may be given during the last hour of the dialysis session to account for vancomycin removal by high flux dialysis mechanisms
4. The 'top up' dose given will depend on the vancomycin level preceding the dialysis session, and the known removal of ~30% by high flux dialysis
5. As a general guide, the 'top up' dose will range from 500mg to 1000mg with consideration of the pre-dialysis vancomycin result

CRRT

Continuous Venovenous Haemodiafiltration (CVVHDF)

Degree of vancomycin removal will be influenced by the filter type and flow rates employed

In general

1. Consider an initial loading dose of 15 to 20mg/kg
2. Subsequent doses of between 1g to 1.5g q24h
3. Trough level to be measured before the 4th or 5th dose

PERITONEAL DIALYSIS

Patients receiving peritoneal dialysis

1. Initial standard loading dose of 2000mg upon starting vancomycin therapy
2. Levels should be repeated approximately every 72 hours to assist with determining when to re-dose
3. A subsequent dose to be given when the vancomycin level falls to $\leq 15\text{mg/L}$
4. In practice, the most common dose prescribed when redosing is 2000mg

Dosage

Vancomycin is suited for either intermittent intravenous administration or continuous intravenous infusion. From a pharmacokinetic and pharmacodynamic perspective vancomycin is ideally suited for intermittent dosing (twice a day dosing).³

The usual susceptibility of common bacterial targets (staphylococci and streptococci) possess low (MIC values of $\leq 1\text{mg/L}$) and the activity of the drug is believed to be concentration-independent.³

Therefore, the likelihood of maintaining free or unbound serum vancomycin concentrations in excess of the bacterial MIC for the entire dosing interval (with standard intermittent dosing) against these target pathogens is usually 100% if the adequate vancomycin trough level is achieved.³

For more information relating to vancomycin drug levels refer to the 'Therapeutic Drug Monitoring' section.

Intermittent dosing

Loading dose (4)

The decision to prescribe a loading dose will generally depend on the severity of the infection, critical nature of the patient and urgency to achieve therapeutic drug concentrations.

A loading dose aims to help the patient reach a therapeutic vancomycin concentration more rapidly (by 12 hours) compared to the absence of a loading dose, however the clinical benefit of this process is not evident.

Overall, the use of a loading dose should be considered for critically unwell patients (eg in ICU, septic, deteriorating clinically) to try mitigate any risk with inadequate vancomycin exposure or delay with achieving therapeutic concentrations.

If a loading dose of vancomycin is considered necessary

- Vancomycin 25 to 30mg/kg (use actual body weight) as a single dose by intravenous infusion

For the management of obese patients at SCGOPHCG, our recommendation is to follow dosing recommendations based on actual body weight in obese patients however a limit or cap for the loading dose should be set to a maximum of 3000mg.

For more information relating to vancomycin dosing in obesity, please refer to the section 'Dosage Adjustments'.

Maintenance dosing

Recommended starting doses (*following a loading dose if loading deemed necessary*) should be based on actual body weight (ABW) with consideration of renal function.

For specific guidance with consideration of both parameters refer to the dosing table below (Table 1)

In obese patients, there is uncertainty relating to the most accurate recommendation for weight-based dosing. There have been reports of increased rates of nephrotoxicity and supratherapeutic vancomycin levels when dosing in obese patients based on ABW.

At SCGOPHCG, our recommendation for dosing in this population is to refer to ABW however a maximum ABW (of 140kg) should be applied to limit dosage.

This mainly applies to vancomycin dosing at initiation to minimise the risk for unnecessary toxicity and excessive plasma concentrations. However, subsequent doses (following achievement of steady state and therapeutic drug monitoring) are excluded from this ABW cap and should be directed by available vancomycin trough concentrations for each patient.

Table 1. Dosing and Monitoring Guide for Initiation of Vancomycin Therapy

Creatinine clearance* (mL/minute)	Suggested dose# ^	Timing for first trough+ concentration level to be taken
Greater than 60mL / min	15 to 20mg/kg 12-hourly	Before the fifth dose
Between 40 to 60 mL / min	15mg/kg 12-hourly	Before the fifth dose
Between 20 to 39mL / min	15 to 20mg/kg daily	Before the fifth dose
Less than 20mL / min	15mg/kg every 48 to 72 hours	Before the second dose is due NB. This level will NOT represent a steady state concentration but instead be useful for guiding when to re-dose the patient
▪ Peritoneal dialysis ▪ Intermittent Haemodialysis ▪ Continuous Renal Replacement Therapy (CRRT)	Refer to section 'Dosing in Specific Populations'	
* Creatinine clearance is generally based on estimates using the Cockcroft-Gault formula		
# Dosages suggested in this table are serve as a guide only for the initial doses prior to trough concentrations becoming available. Many other influencing factors are not taken into consideration in the above guide. Clinical assessment and review of the patient should accompany the decision for the dose of vancomycin to prescribe. The trough concentration result should be used to help guide subsequent doses.		
^ Doses should be based on actual body weight (ABW) but for morbidly obese patients The ABW should be capped at 140kg, with consideration to calculate doses at the lower end of the dosage range in these cases.		
+ Trough levels should be collected 30 to 60 minutes before the next dose The most reliable vancomycin level for interpretation purposes should be a trough level that is collected following attainment of steady state concentrations in the body (eg after approximately 5 half-lives). For more information refer to section 'Therapeutic Drug Monitoring' The first trough level may be collected before the fourth dose in patients who have received a loading dose of vancomycin Following the collection of a vancomycin trough level, do NOT wait for the result to return before giving the next dose UNLESS specifically advised or in severe renal impairment		

Preparation (5)

Dissolve each 1g vial of vancomycin using 20mL of sterile water for injections (WFI). The required amount of vancomycin should then be diluted further by adding to the correct volume of a compatible fluid (eg sodium chloride 0.9%, glucose 5%) according to the 'Administration guide – Table 2' below

Concentration

The final concentration of the solution to be administered intravenously should contain approximately 5 to 10mg/mL of vancomycin (following dilution in the compatible fluid).

The maximum concentration of vancomycin (even in fluid restricted individuals) should not exceed 10mg/mL.

Compatible fluids

Water for injections (WFI), sodium chloride 0.9%, glucose 5%, glucose 10%, Hartmann's solution, Plasma-Lyte 148

Administration (5)

- Vancomycin must be administered as an intravenous infusion at an appropriate rate based on the dose prescribed (refer to the Administration guide - Table 2 for guidance)
- Do NOT administer vancomycin as an intramuscular or subcutaneous injection as this may lead to skin ulceration and tissue necrosis since vancomycin is a highly irritant drug.
- Follow recommendations pertaining to dilution volumes and infusion rates as per the Administration table to avoid infusion related reactions
- Record the time of administration. This is very important for the interpretation of a plasma trough concentration

Administration rates advice (5)

- Please refer to Table 2 for recommendations for the rate of infusion based on the dose of vancomycin and intravenous access available
- It is important to note that the recommended concentrations and rate of infusion allowed may differ slightly between administering through a central venous access device (eg CVC, PICC) and a peripheral intravenous cannula (PIVC)
- Thrombophlebitis is another complication that may arise from the administration of more concentrated infusions than those recommended in the table, especially if administering via an intravenous peripheral cannula.
- Vancomycin infusion reaction (VIR) is a well-recognised reaction that may occur following administration of vancomycin, especially following rapid infusion. The maximum rate of administration of vancomycin via a PIVC should NOT exceed 10mg/minute. The recommended rates of infusion provided in the table may help reduce the risk of VIR or hypersensitivity.
- In patients who have experienced a previous reaction following vancomycin administration (eg flushing, thrombophlebitis etc) it is recommended to prescribe a slower infusion rate or even less concentrated solution to assist with minimising the recurrence or severity of any infusion-related reaction.

More information on 'Vancomycin Infusion Reaction' see under the 'Adverse Effects' section.

Table 2. Vancomycin Intravenous Administration Guide

Table 2: Vancomycin Intravenous Administration Guide			
Type of parenteral access available	Dose	Minimum Volume	Recommended Rate
Peripheral intravenous cannula (PIVC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes (eg 1 hour)
	1.25 to 1.5g	≥ 250mL	Infuse over at least 90 minutes (eg 1.5 hours)
	1.75 to 2g	≥ 250mL (preferably 500mL)	Infuse over at least 120 minutes (eg 2 hours)
	2.25 to 2.5g	≥ 500mL	Infuse over at least 150 minutes (eg 2.5 hours)
	2.75 to 3g	≥ 500mL	Infuse over at least 180 minutes (eg 3 hours) (extend to 240 minutes if not tolerated)
	>3g	Refer to your clinical pharmacist for advice	
Central venous access device (eg CVC, PICC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes (eg 1 hour)
	1.25 to 1.5g	≥ 250mL (≥ 150mL via burette)	Infuse over at least 90 minutes (eg 1.5 hours)
	1.75 to 2g	≥ 250mL (≥ 200mL via burette)	Infuse over at least 120 minutes (eg 2 hours)
	2.25 to 2.5g	≥ 250mL	Infuse over at least 150 minutes (eg 2.5 hours)
	2.75 to 3g	≥ 500mL	Infuse over at least 180 minutes (eg 3 hours)
	>3g	Refer to your clinical pharmacist for advice	
Nb. If your patient experiences an infusion-related reaction then reduce the rate of the infusion immediately and consider reducing the concentration for the subsequent dose, depending on the nature of the reaction. This will only apply if deemed clinically safe to continue using vancomycin. For severe reactions, vancomycin may need to be ceased immediately and substituted for an alternative antibiotic. Contact an ID staff member for advice on alternative options. Contact your clinical pharmacist for advice on adjustments to the infusion rate or dilution.			
Intravenous injection (bolus dose)	Not recommended – rapid bolus injections of vancomycin can cause severe infusion-related reactions involving flushing, pruritus, dyspnoea and even hypotension. All intravenous vancomycin doses should be administered as an infusion as per the recommendations within this table.		
Intramuscular injection	Contraindicated – may cause ulceration and tissue necrosis		
Subcutaneous injection	Contraindicated – may cause ulceration and tissue necrosis		
Requests for a vancomycin continuous infusion should be referred to your clinical pharmacist for advice			

Quick Reference Pearls**Recommended vancomycin concentrations**

- **Via PIVC** 5mg/mL (more concentrated solutions have been tolerated and may be used if considered necessary)
- **Via CVAD** 5 to 10mg/mL

Recommended vancomycin infusion rates

- **Via PIVC** 10mg/minute (may be infused at faster rate, up to 16.66mg/minute if tolerated or considered necessary)
- **Via CVAD** 16.66mg/minute

Dosage Adjustments and Recommendations in Specific Populations

Renal impairment (acute or chronic)

Vancomycin is eliminated primarily by the kidneys with > 80-90% of the drug recovered unchanged in the urine within 24 hours of a single dose.

Therefore, vancomycin dosage adjustments are primarily based on the patient's renal function.

Dosing in renal impairment will vary depending on many factors including dialysis, patient weight, indication, location of infection etc.

The half-life of vancomycin is extended significantly due to renal insufficiency, which directly prolongs the time to reach steady state levels in the body.

To account for this, a loading dose for renally impaired patients is recommended to expedite the process.

The subsequent maintenance therapy doses should be according to the renal function as per Table 1 above.

Please remember that the dosing advice in the table serves only as a guide for initial doses of vancomycin and other factors as mentioned above need to be considered.

For subsequent doses, trough serum concentration monitoring will be the most accurate and reliable measurement to further guide dosing, especially in patients with renal dysfunction (see below).

Advanced Renal Impairment (dialysis dependent) (6, 7)

Peritoneal dialysis (6, 7)

- Vancomycin is NOT removed by peritoneal dialysis
- Dosing recommendations
 - Initial standard loading dose of 2000mg is given
 - The first trough level should be collected approximately 72 hours following the loading dose
 - A subsequent dose should only be administered when the plasma vancomycin concentration falls to $\leq 15\text{mg/L}$

The subsequent dose should be guided by the individual patient parameters (eg body size, severity of infection, type of infection, pathogen etc) however, as a reference the dose commonly prescribed is 2000mg

- Repeat vancomycin levels approximately every 72 hours and administer another dose when the level falls to $\leq 15\text{mg/L}$.

Intermittent Haemodialysis (low flux) (19)

- This form of dialysis is not commonly used at SCGH.
- Vancomycin is only minimally removed by low flux haemodialysis due to its medium-sized molecular weight and lower porosity of the low flux dialysis membrane
- Vancomycin dosing should be based on a creatinine clearance of $< 20\text{mL / min}$ as per dosing table
- Subsequent doses should be guided by vancomycin plasma levels.

Intermittent Haemodialysis (high flux) (6, 7, 8, 19)

- This is the **main form of dialysis used at SCGH**
- Vancomycin is partly removed by high flux haemodialysis. The average amount removed by high flux haemodialysis during a 3 to 4hour dialysis session is between 30 to 38%.
- Dosing recommendations
 - Dosing pre- and during each dialysis session
 - Initial loading dose of 15 to 20mg/kg based on ABW upon initiation of vancomycin therapy.
 - First vancomycin level collected approximately 48 to 72 hours following the loading dose.
 - The vancomycin levels should be aligned with the timing of the next dialysis session (eg before the HD session), as this level preceding the dialysis session will be used to guide any vancomycin 'top up' doses during the dialysis session.
 - During the last hour of the high flux haemodialysis session, depending on the vancomycin level that returned earlier in the day an additional 'top up' dose will be administered during the last hour of the dialysis session to account for the expected removal of vancomycin by the high flux dialysis (10)
 - This additional dose will depend on a few factors, including the vancomycin level preceding the high flux dialysis session and the degree of vancomycin removal by high flux HD (~ 30%)
 - As a guide, the 'top up' dose will range from 500mg to 1000mg, with consideration of the factors discussed above.

Haemodiafiltration (HDF) (9, 10)

- An alternative form of dialysis to standard intermittent high flux haemodialysis where convective clearance methods are incorporated to assist with removal of middle and large sized compounds such as uraemic toxins and drug molecules.
- Additional removal of vancomycin compared to high flux HD is anticipated due to the convection clearance component provided with HDF
- Would suggest a similar regimen to dosing in high flux HD above, with a slightly higher dose to account for the greater removal by HDF during each dialysis session
- Dosage recommendations
 - Dosing pre- and during each dialysis session
 - Initial loading dose of 15 to 20mg/kg based on ABW upon initiation of vancomycin therapy
 - Repeat doses when the plasma vancomycin level is $\leq 15\text{mg/L}$
 - During each HDF session, an empiric 750mg should be administered during the last hour of the HDF dialysis session to account for expected vancomycin removal during the session.

Continuous Arteriovenous Haemofiltration (CAVH)**Continuous Venovenous Haemofiltration (CVVH)****Continuous Arteriovenous Haemodialysis (CAVHD)**

- Vancomycin is removed by these forms of dialysis
- Dosing recommendation
 - 1g every 48 hours (6)

Continuous Venovenous Haemodiafiltration (CVVHDF)**Continuous Venovenous Haemodialysis (CVVHD)**

- Vancomycin is effectively removed by CVVHDF and CVVHD (both used commonly in ICU)
- Removal by these forms of dialysis is improved due to the use of diffusion gradients but CVVHDF uses additional convective solute transport methods.
- Removal is however highly dependent on the filter type and flow rates employed (8)
- Elimination by CVVHDF accounts for nearly 50% of total vancomycin clearance in patients receiving CVVHDF with the remainder removed by non-renal clearance mechanisms.
- Dosing recommendations for patients receiving CVVHDF and CVVHD
 - An initial loading dose of 15 to 20mg/kg
 - Subsequent maintenance dosage should be 1g to 1.5g q24h (11)
- Recommend a trough level for monitoring before the fourth dose further to guide dosing

Hepatic impairment (17)

- There are no specific guidelines advising dosage adjustments for vancomycin in hepatic impairment
- Studies have demonstrated that varying degrees of hepatic dysfunction does not appear to alter the pharmacokinetics of vancomycin so dosage adjustments in this cohort of patients is not required

Obesity (3, 13)

- There isn't a clear consensus regarding the preferred body size descriptor to be employed with vancomycin dosing in obese patients.
- This is a growing concern as statistics obtained from the Center for Disease Control and Prevention (CDC) have revealed that the prevalence of obesity in the United States has increased from 30.5% (1999-2000) to 42.4% (2017-2018).
- At SCGOPHCY, our recommendation is to dose vancomycin on ABW with a maximum limit set to 140kg.
- For dosing purposes, two of the most recognised determinants of dosage calculations for vancomycin relate to its clearance and volume of distribution (VoD).

Vancomycin is a hydrophilic drug and pharmacokinetic studies have demonstrated that the volume of distribution of vancomycin increases with increased actual body weight, however not in a proportional manner.

Most available literature demonstrate a linear correlation between the clearance of vancomycin with actual body weight.

Overall, there is conflicting evidence in the currently available data pertaining to this aspect of vancomycin dosing and therefore no clear consensus exists. Further studies are required.

Elderly patients (14)

- The total body clearance of vancomycin is reduced in elderly patients (17)
- Caution should be taken when prescribing doses for the elderly, especially in elderly patients with impaired renal function of low body weight.

Neutropenic patients (14)

- A pharmacokinetic study assessing vancomycin clearance showed that neutropenic patients cleared the drug at a higher rate when compared with non-neutropenic patients

Therapeutic Drug Monitoring

Background

Vancomycin drug monitoring is an essential aspect of the prescribing and use of this medication, due to the potential toxicities associated with its use, and also the correlation between target drug levels and therapeutic efficacy.

Vancomycin monitoring is a very complex aspect of its use.

Recent consensus guidelines and publications have shifted towards direct PK/PD (pharmacokinetic/pharmacodynamic) efficacy target measurements of area under the curve (eg AUC/MIC) instead of the routine serum trough concentration monitoring, by aid of new pharmacokinetic Bayesian software programs.

However, SCGOPHCG have not transitioned to this change in monitoring recommendations yet and currently still follow routine targeted trough measurements of vancomycin levels, whilst awaiting further evidence regarding this practice change to emerge.

Routine therapeutic monitoring of vancomycin is centred on achieving trough concentrations of vancomycin between 15 and 20mg/L (with acceptance of higher levels towards 25mg/L).

The rationale for this target range is based on the clinically based assumption that trough concentrations within this range are a surrogate marker for achieving the target AUC/MIC ratio of ≥ 400 (given the historical difficulty with estimating AUC previously) for organisms with an MIC ≤ 1 mg/L.

Currently, an AUC/MIC ratio of ≥ 400 is considered the optimal PK/PD efficacy target (based on multiple retrospective single centre observational studies of MRSA bloodstream infections).

Trough concentration monitoring remains a practical alternative to AUC/MIC guided dosing in clinical practice where the Bayesian PK software is unavailable.

Trough concentration attainment ensures achievement of a minimum cumulative exposure however its limitations lie in the fact that the trough only represents a single exposure point at the end of the dosing interval (eg a single minimum daily AUC value only) whereas 24 hour AUC monitoring will provide the average concentration or exposure to the drug over the entire dosing time period.

There can be considerable variability in the upper range of AUC values associated with each given trough value. Therefore, despite trough monitoring being a practical approach to vancomycin TDM, it is important to note that it may be insufficient to reliably guide vancomycin therapy for all patients.

Recommendations (3, 16)

- Vancomycin therapeutic monitoring is **strongly recommended** for all patients receiving the therapy for > 48 hours.
- Monitoring of **plasma trough concentrations** is highly recommended to help **minimise** the risk for **nephrotoxicity** (induced by supratherapeutic concentrations of vancomycin) and potential **therapeutic failure** secondary subtherapeutic vancomycin levels.
- Vancomycin is eliminated mainly by the renal system; via glomerular filtration, therefore any decrease in the glomerular filtration rate may lead to an increase in serum vancomycin concentration.
 - There is an association between vancomycin TDM and a reduction in incidence of nephrotoxicity

Vancomycin level information

- The type of vancomycin level to be collected should be a serum trough concentration (eg a level collected just before the next dose is due).
- The first vancomycin level should only be collected after steady state concentration of the drug has been reached.
- This is generally achieved after 3 to 5 half-lives of the drug.
- For vancomycin, our recommendation is to collect the first trough level only after 4 or 5 doses has been administered (eg level just before the 5th dose in a patient with normal renal function)

Details for the timing of the first vancomycin trough level are detailed within the dosing table above.

- *For patients with advanced renal impairment (eg requiring dialysis support), the timing of the levels will vary slightly and due to the extended drug half-life in these patients the initial levels collected are not expected to be at steady state concentration. However, these levels still serve a purpose in guiding the timing and suitability for when the next dose should be given.*
- If there are any vancomycin dosage adjustments made, a repeat level should NOT be performed until 48 hours following the dosage change to allow for re-establishment of steady state concentrations. Again, this recommendation may not directly apply to patients with chronic advanced renal impairment receiving dialysis therapy due to a change in pharmacokinetics and other factors contributing to drug removal.

Vancomycin level sampling / collection

- Vancomycin levels should be collected as a trough sample (eg approximately 30-60 minutes before the next dose)
- Each level must be collected from a peripheral site (eg not from any central access device that is already in situ)
- The blood sample should be collected into either a 'DARK GREEN top' or 'RED top' tube as per PathWest guidelines. The dark green top tube is preferred when collecting samples from within the metropolitan hospitals.



- The timing of the sample collection should be clearly documented on the pathology request form and on the medication chart (only if there is room for this to be entered).
- Following collection of a sample for vancomycin concentration testing, the next dose of vancomycin should be administered as per usual (eg do not wait for the result to return before administering the next dose).
- We do NOT recommend withholding any subsequent vancomycin doses whilst awaiting a routine level to return, unless specifically advised by the medical prescriber or clinical pharmacist.

Vancomycin target trough level range

- The accepted target serum vancomycin trough concentrations are between **15 to 20mg/L** (however levels up to 25mg/L are accepted and shown to be safe, especially in certain clinical situations involving complex infections or resistant organisms).
- In general, the target concentrations may vary depending on the specific pathogen involved and specific indication.
- For more advice regarding the most appropriate target vancomycin trough concentration contact an ID Pharmacist or ID Physician.

Adjusting the dose of vancomycin based on therapeutic drug monitoring

The following recommendations serve as a guide only.

Clinical assessment and review of other factors contributing to the vancomycin level must be considered in addition to referencing this table (eg timing of level, evolving renal dysfunction, interacting medications)

Advice within this table does NOT apply to patients with known chronic renal impairment receiving dialysis

Important to first review that each vancomycin trough level result is suitable for interpretation

- Is it an accurate trough level, collected at the correct time?
- Was the trough level collected when the drug had reached steady state concentration in the patient? Eg collected only after an adequate number of doses have been received first

Table 3. Dosage Adjustment Guide for Vancomycin

Vancomycin trough concentration (mg/L)	Recommended dose adjustment for next dose (to achieve target vancomycin trough concentration 15 to 25mg/L)
>40	1. Withhold the next dose 2. Repeat another vancomycin level before the next scheduled dose is due (eg in 12-24hrs) 3. Investigate for contributing factors or potential causes of the high vancomycin level (eg renal dysfunction, concurrent nephrotoxins) 4. If the repeat vancomycin level remains supratherapeutic or renal function has significantly deteriorated, consider ceasing vancomycin therapy and contact ID for advice on potential alternative treatment options 5. If the repeat vancomycin level returns within target range, without evolving renal dysfunction then contact your clinical or ID pharmacist for dosing guidance if vancomycin therapy continues
>30	1. Withhold the next dose 2. Repeat another vancomycin level before the next scheduled dose is due (eg 10 hours later) 3. If level returns within or below the target range then proceed with the next dose but adjust dose accordingly first (suggest <u>reducing dose by approximately 1/3 to 1/4 of original dose</u>, pending the level that has returned) 4. If the level returns and is still elevated (eg above the target range) then continue to withhold vancomycin doses and investigate reason/s for persistently elevated vancomycin concentrations. Consider ceasing vancomycin therapy and contact ID to discuss potential alternative treatment options instead
25 to 30	1. Review renal function / signs of evolving renal dysfunction 2. If no evidence of developing or established renal dysfunction then continue with next scheduled dose but adjust by reducing by 1/3 of the original dose (eg usually 250 to 500mg) 3. Repeat another vancomycin level after 48 hours and continue to closely monitor renal function
15 to 24 (target range)	No change required. Continue current dose.
10 to 14	1. If treating an uncomplicated infection and patient is clinical responding/stable then continue with the current dose. 2. If treating a complex, deep-seated infection and patient is not clinically improving then consider <u>increasing the total daily dose by 500mg</u>
< 10	1. <u>Increase</u> the <u>total daily</u> vancomycin dose by 750 to 1000mg 2. Repeat another vancomycin trough level following 4-5 doses 3. If persistently <10mg/L, contact the ID Pharmacist for advice. Potential options may include re-loading the patient with vancomycin or increasing the total daily dose by further 1000mg.

Nb. The recommendations within this table serve only as a guide to assist with dosage adjustments for vancomycin. Many other influencing factors are not completely accounted for by this guidance table, and therefore the clinical judgement of the treating staff must take precedence and be considered in all cases.

For further information contact your Clinical Pharmacist.

Adverse effects (1, 2, 8, 14, 18)

- Infusion related reactions (eg Vancomycin Infusion Reaction, previously referred to as 'Red man syndrome')
 - Involves flushing of the upper body, pain and muscle spasms of the chest and back
 - Related to rapid infusion of vancomycin
 - See dedicated section on this reaction (Vancomycin Infusion Reaction - VIR) below
- Anaphylactoid reactions – hypotension, heart palpitations, tachycardia, wheezing, dyspnoea, urticaria, pruritus
 - This reaction may develop during or soon after commencement of vancomycin infusion and usually resolves within 20 minutes however may persist for several hours
 - Severe cases may require immediate treatment with adrenaline, corticosteroids and supplemental oxygen
- Sensorineural deafness and tinnitus
 - Generally reversible deafness experienced in patients with normal baseline auditory and renal function
 - Permanent deafness more likely to occur in patients with existing compromised auditory or renal function
- Vertigo and dizziness
- Tissue irritation and necrosis (usually following intramuscular injection or extravasation from intravenous dosing)
- Neutropenia
 - Generally reversible upon cessation of therapy
 - Usually observed > 1 week following commencement of therapy and associated with a total cumulative dose of $\geq 25\text{g}$
- Thrombocytopenia
 - Generally reversible upon cessation of therapy
- Renal dysfunction
 - Manifested by an increase in serum creatinine or urea concentrations
- Interstitial nephritis (rare)

Vancomycin infusion reaction (VIR) (2, 18)

- This reaction is a type of adverse reaction, previously referred to as 'red man syndrome'
- This reaction is not a true allergic reaction, but rather a rate-dependent infusion reaction
- It may be difficult to distinguish between VIR and anaphylaxis as both reactions involve flushing and hypotension however hives, laryngeal oedema and wheeze are more suggestive of an anaphylactic reaction.

Treatment and management of anaphylaxis is different from the treatment of Vancomycin Infusion Reactions
If there is any concern for **anaphylaxis** or a more serious drug reaction, then immediate medical attention is required and should be managed as an **urgent medical emergency**

Cause

- A rate-related infusion reaction triggered by rapid infusions of vancomycin by intravenous route

Signs and symptoms

- Characterised by dramatic erythema, flushing and pruritus affecting the upper body, neck and face (upper body affected more than lower body)
 - Pain and muscle spasms in the back and chest
 - Dyspnoea
 - Hypotension
- VIR is rarely life threatening, however can cause severe cardiovascular toxicity and even cardiac arrest
- Important to note that immunoglobulin E (IgE) mediated anaphylaxis and present with similar symptoms to VIR, however treatment of each diagnosis differs
- VIR usually occurs with initial dose of vancomycin
- IgE mediated reactions do NOT occur with the initial dose as it requires sensitisation from a prior dose first

Mechanism

- VIR is an idiopathic reaction; an adverse drug reaction with signs and symptoms that mimic immunologic drug allergies without IgE mediated mechanisms
- Vancomycin can directly activate mast cells to release vasoactive mediators (eg histamine) which trigger the infusion reaction and symptoms in the patient

Management

- The recommended approach to managing VIR should be based on the severity of the reaction
- Mild reactions (flushing only, not discomfort to the patient)
 - No specific treatment required
 - Symptoms usually resolve in minutes after temporarily ceasing treatment
 - Can re-commence the infusion at a slower rate (eg halve the rate from previous, or at $\leq 5\text{mg/minute}$).
- Moderate reactions (flushing symptoms associated with pruritus and causing general discomfort, however haemodynamically stable with NO chest pain or muscle spasms)
 - Interrupt and pause the infusion
 - Give the patient an antihistamine (eg oral loratadine, IV/oral famotidine).
 - Oral and intravenous antihistamine premedications appear to be similarly efficacious
 - Symptoms should resolve shortly after
 - The infusion can then be restarted at half the previous administration rate, or even slower (eg $\leq 5\text{mg/minute}$)
 - The slower infusion rate should be prescribed for subsequent infusions with consideration of premedication therapy
- Severe reactions (previous symptoms above but also involving muscle spasms, chest pain or hypotension)
 - Stop the infusion immediately
 - Give the patient an antihistamine. Recommendations are the same as above, however there are some studies that support the use of IV formulations including IV famotidine, which is an SAS medication and IV diphenhydramine (not available in Australia). As an alternative, if an IV formulation is required the IV promethazine may be considered.
 - After resolution of symptoms, the infusion can be reintroduced at a much slower infusion rate. The recommended rate to administer the vancomycin over would be at least 4 hours.
 - If tolerated at this rate, future doses should then be infused at the same reduced rate (eg 4 hours or longer) with the use of premedication (eg antihistamines) before each dose.

*****IMPORTANT*****

If anaphylaxis is suspected (*eg hives, laryngeal oedema, hoarse voice, wheezing, breathing difficulty, tachycardia, hypotension, vomiting, dizziness etc*) or there has been NO clinical response to the use of antihistamines for the symptoms experienced then the recommendations above will NOT apply for future doses.

Reducing the rate of infusion and the use of antihistamine premedications will NOT prevent IgE-mediated anaphylaxis reactions.

In cases of confirmed or likely anaphylaxis to vancomycin, an alternative antibiotic should be used instead.

Pharmacokinetic information (14)

Absorption

- Oral bioavailability: negligible. Poorly absorbed after oral administration. Slight increase in absorption may appear in conditions causing inflammation of the GIT (eg inflammatory bowel disease, pseudomembranous colitis)
- Intraperitoneal bioavailability: ranges from 38 to 60%. Bioavailability increases with increased dwell time (eg up to 60% of a dose may be absorbed into systemic circulation over 6hr period following intraperitoneal administration (8))
- Ocular bioavailability: may reach therapeutic concentrations in the anterior chamber of the eye when administered as a topical eyedrop at a concentration of 50mg/mL. The average concentration of vancomycin in the anterior chamber following conventional method of eyedrop delivery was 2.04 microg/mL.

Distribution

- Distributes widely into body tissues and fluids except for the CSF (when uninflamed) (10, 17, 18)
- Volume of distribution ranges from 0.2 to 1.55 L/kg (7, 10)
- Protein binding ranges from 30 to 55% (7, 17)
- Pharmacokinetics are best described by a two or three compartmental model (17)
 - Short distribution phase
 - Intermediate distribution phase (serum concentration decline when the drug travels from the blood to the tissues)
 - Variable elimination phase – a slower fall in vancomycin concentrations based on the elimination rate constant with influence from renal function
- Steady state concentrations are achieved generally after 4 to 5 half-lives of the drug

Penetration

- Penetrates into majority of body spaces, although the concentrations are variable and are often dependent on the degree of inflammation present (eg inflammation improved vancomycin penetration) (15)
- Penetration into the CSF is minimal in the absence of meningeal inflammation (17). This is also further reduced by concurrent administration of dexamethasone).
- Penetration into the lungs is highly variable and ranges from 20 to 30% of serum concentration only (15, 17)
- Animal studies have demonstrated good penetration (high concentrations) in the kidney, liver, aorta, heart tissue and associated abscess fluids of those areas.
 - The vancomycin concentrations in heart valve tissue, subcutaneous tissue and muscle were 52%, 29% and 27% of the serum concentration respectively.

- Penetration into cardiac vegetations (eg infective endocarditis) has been demonstrated in experimental models.
- Vancomycin levels achieved in pericardial, ascitic, pleural and synovial fluids have been considered adequate and reaching therapeutic concentrations (17)
- Appears to penetrate bile fluid however does not appear to effectively maintain adequate therapeutic concentrations (17). Its effectiveness in cholangitis as sole therapy without source control may be questionable.
- Achieves relatively high concentrations in breast milk (17)
- Undergoes transplacental passage during the second trimester of pregnancy (17)
- Diffuses readily into urine and peritoneal dialysis fluid to reach minimum inhibitory concentrations of most pathogens (17)

Metabolism

- No apparent metabolic pathway. Does not appear to be metabolised to any significant degree (16)
- Excreted primarily unchanged (> 80%)

Excretion

- Primarily excreted unchanged via the kidneys by glomerular filtration (17)
- No evidence of tubular secretion or resorption (17)
- Renal mechanisms account for >90% of vancomycin clearance (19)
- Non renal clearance accounts for only <5%. Therefore, drug clearance and dosing is highly dependent on the patient's renal function
- Reports of lower vancomycin clearance in hepatic dysfunction however this is unlikely to be of clinical significance given the majority of vancomycin clearance is via the renal pathway (17)

Elimination half-life

- Ranges from 4 to 6 hours (will be prolonged in renal impairment) (16)
- In anephric patients the drug half-life extends to 7.5 days (16)

Practical points

Drug interactions (1, 8, 14)

- Neurotoxic or nephrotoxic agents (eg aminoglycosides; gentamicin, amikacin, tobramycin, piperacillin-tazobactam, amphotericin B, colistin, cisplatin, ciclosporin, lithium) (1)
- Loop diuretics (eg furosemide, bumetanide, ethacrynic acid) - caution with use as this combination may aggravate ototoxicity (1)
- Suxamethonium/vecuronium/rocuronium and other neuromuscular blockers – combination with vancomycin may enhance neuromuscular blocking properties (16)

IV Drug Compatibility data (5)

Compatible fluids

- Glucose 5%, glucose 10%, Hartmann's solution, sodium chloride 0.9%

Compatible drugs via Y-site

- Aciclovir, amifostine, amiodarone, anidulafungin, aztreonam, buprenorphine, calcium chloride, calcium gluconate, caspofungin, cefuroxime, ciclosporin, clindamycin, dexamethasone, digoxin, ephedrine sulfate, erythromycin, esmolol, fentanyl, filgrastim, fluconazole, gentamicin, granisetron, hydromorphone, insulin (NovoRapid), isavuconazole, labetalol, lidocaine, linezolid, magnesium sulfate, metoclopramide, midazolam, morphine sulfate, mycophenolate mofetil, nifedipine, palonosetron, paracetamol, pethidine, posaconazole, remifentanyl, tigecycline, tobramycin, zidovudine
- Piperacillin-tazobactam => concentration dependent. Concentration of vancomycin up to 5mg/mL in NaCl 0.9% are compatible with piperacillin-tazobactam (up to 90mg/mL of piperacillin) in NaCl 0.9%, however higher concentrations are incompatible.

Incompatible fluids

- Albumin

Incompatible drugs

- Albumin, aminophylline, azathioprine, bivalirudin, calcium folinate, daptomycin, defibrotide, epoetin alfa, flucloxacillin, foscarnet, furosemide, ganciclovir, ketorolac, methylprednisolone, moxifloxacin, omeprazole, rocuronium, sodium bicarbonate, sodium valproate, urokinase
- Vancomycin is also incompatible with many beta-lactam antibiotics (eg penicillins, cephalosporins, carbapenems). If infusing together with high concentrations of vancomycin then precipitation is more likely to occur. Flush the line well between doses.

Pregnancy & Lactation advice

ADEC Pregnancy Category B2 (2, 5)

- It is unknown whether vancomycin administered to pregnant women will cause foetal harm. Only use in pregnant women if clearly needed and no suitable alternatives exist.
- Categorised as drugs which have been used by only a limited number of pregnant women and women of childbearing age, without an increase in the frequency of malformations or other direct or indirect harmful effects on the human foetus having been observed. Studies in animals are inadequate.

Safe in lactation

- Categorised as generally compatible for use in breastfeeding (2, 5)
- Vancomycin is excreted in breast milk however it is unknown whether it causes any harm to the newborn. Caution should still be exercised when using in nursing mothers.
- May induce loose bowel actions in the infant

Storage (1, 5)

- Store vials below 25°C
- Protect vials from light
- Once reconstituted, the solution is stable for only 24 hours at between 2 to 8°C

Acknowledgements

- Vancomycin Dosing and Monitoring Guidelines (Appendix 1) – Royal Brisbane and Women's Hospital Intensive Care Unit

Related Documents

Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

Authorising Policy and Standard/s:

- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [SCGOPHCG Formulary Management Procedures MMP013](#)
- [SCGH Hospital Policy HP261 High Risk Medications](#)
- [SCGH Hospital Policy 141 Medicines Management](#)
- [OPH Hospital Policy High Risk Medication](#)
- [Clinical Governance, Safety and Quality Policy Framework](#)
- NSQHS Standard 1.1.1: An organisation-wide management system is in place for the development, implementation and regular review of policies, procedures and/or protocols.
- NSQHS Standard 1.7: Developing and/or applying clinical guidelines or pathways that supported by best available evidence (all subsequent actions of this criterion are applicable).
- NSQHS Standard 4.1: Developing and implementing governance arrangements and organisational policies, procedures and /or protocols for medication safety, which are consistent with national and jurisdictional legislative requirements, policies and guidelines.
- NSQHS Standard 4.1.2: Policies, procedures and/or protocols are in place that are consistent with legislative, national, jurisdictional and professional guidelines.

Standards, procedures, guidelines:

- Australian Commission on Safety and Quality in Healthcare. National terminology, abbreviations and symbols to be used in the prescribing and administering of medicines in Australian hospitals. 2008.
- [OD 0385/12](#). National Recommendations for User-Applied labelling of Injectable Medicines, Fluids and Lines. Issued 2012.
- <http://www.safetyandquality.gov.au/internet/safety/publishing.nsf/Content/D0DABD9912D44A14C>
- [Intravenous Therapy](#). SCGH Nursing Practice Guideline 4. Reviewed 2013.
- [Drug Administration](#). SCGH Nursing Practice Guideline 51. Reviewed 2013.
- [Drug Administration](#). SCGH Hospital Policy 141. Reviewed 2013.
- [Prescribing and Drug Supply for Inpatients](#). SCGH Hospital Policy 197. Reviewed 2011.
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For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.

VANCOMYCIN QUICK REFERENCE GUIDE

HIGH RISK MEDICATION

This is a high-risk medication. Strict monitoring and administration requirements apply to the use of the medication

DRUG	PRESENTATION	DOSAGE	DILUTION VOLUME	ADMINISTRATION RATE	THERAPEUTIC DRUG MONITORING
Vancomycin	Glass vial containing lyophilised powder (white to light brown)	LOADING DOSE If deemed necessary 25 to 30mg/kg (ABW) as single dose Max 3000mg	The final concentration of the solution should be 5 to 10mg/mL Final concentration should NOT exceed 10mg/mL, even in fluid restricted individuals	PERIPHERAL ACCESS Recommend 10mg/min May be increased up to 16.66mg/min but only if necessary and patient tolerating CENTRAL ACCESS Recommend 10mg/min with option to increase to 16.66mg/min See administration guide below	Collect trough levels after steady state concentration has been reached (eg after 4-5 doses have been given) Target level 15-20mg/L (up to 25mg/L) Collect sample using DARK GREEN or RED top  DARK GREEN (DKGNLITH) Lithium Heparin [No Gel]  RED Serum [No Gel] Following sampling, do NOT wait for the result to return before giving next dose unless specifically advised
	PREPARATION Dissolve each 1g vial using ~ 20mL WFI. This must be further diluted See administration guide below	MAINTENANCE Dependent on renal function See dosing table below			

DOSING GUIDE (SUMMARISED)

CrCl* (mL/minute)	Suggested dose# ^	Timing for first trough* concentration level collection
Greater than 60mL / min	15 to 20mg/kg 12-hourly	Before the fifth dose
Between 40 to 60 mL /min	15mg/kg 12-hourly	Before the fifth dose
Between 20 to 39mL /min	15 to 20mg/kg daily	Before the fifth dose
Less than 20mL / min	15mg/kg every 48 to 72 hours	Before the second dose is due NB. This level is used for guiding when to re-dose
* Creatinine clearance is generally based on estimates using the Cockcroft-Gault formula		
# Dosages suggested in this table are serve as a guide only for the initial doses prior to trough concentrations becoming available. Many other influencing factors are not taken into consideration in the above guide. Use ABW for dosage calculations		

ADMINISTRATION GUIDE (SUMMARISED)

Type of parenteral access available	Dose	Minimum Volume	Recommended Rate
Peripheral intravenous cannula (PIVC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes
	1.25g to 1.5g	≥ 250mL	Infuse over at least 90 minutes
	2g	500mL	Infuse over at least 120 minutes
Central venous access device (eg CVC, PICC)	≤ 1g	≥ 100mL	Infuse over at least 60 minutes
	1.25g to 1.5g	≥ 250mL	Infuse over at least 90 minutes
	2g	≥ 250mL	Infuse over at least 120 minutes
AVOID SUBCUTANEOUS, INTRAMUSCULAR OR BOLUS IV INJECTION			

DOSE ADJUSTMENT GUIDE (SUMMARISED)

Vancomycin trough level (mg/L)	Recommended dose adjustment for next dose (to achieve target vancomycin level 15 to 25mg/L)
>40	Withhold the next dose and repeat level before timing of next scheduled dose Investigate for contributing factors/potential causes for the high vancomycin level Refer to the guideline section 'vancomycin dose adjustment guide' for advice
>30	Withhold the next dose and repeat level before timing of next scheduled dose Refer to the guideline section 'vancomycin dose adjustment guide' for advice
25 to 30	Review renal function / signs of evolving renal dysfunction If no evidence of developing renal dysfunction then continue with next scheduled dose but reduce by 1/3 of the original dose (eg usually 250 to 500mg)
15 to 24 (target range)	No change required. Continue current dose.
10 to 14	If target level is 20-25mg/L, consider increasing the total daily dose by 500mg
< 10	Increase the total daily vancomycin dose by 750 to 1000mg

If your patient experiences an **infusion-related reaction**, reduce the rate of the infusion and refer to the section on 'Management of Vancomycin-infusion reactions'

If your patient experiences a **severe reaction such as anaphylaxis**, STOP the infusion immediately and seek medical help

VANCOMYCIN QUICK REFERENCE GUIDE

HIGH RISK MEDICATION

This is a high-risk medication. Strict monitoring and administration requirements apply to the use of the medication

DOSING IN ADVANCED RENAL IMPAIRMENT (DIALYSIS)

SEVERE ACUTE RENAL IMPAIRMENT

Severe renal impairment but not requiring dialysis

1. Consider the use of a loading dose (25 to 30mg/kg) when starting vancomycin therapy
2. A subsequent dose of 15mg/kg to be given only when the vancomycin level falls to $\leq 15\text{mg/L}$
3. Perform repeat plasma vancomycin levels every 24 to 48 hours to determine timing for redosing

Based on clinical experience the doses will be approximately every 48 to 72 hours

HAEMODIALYSIS (HIGH FLUX)

Patients receiving high flux haemodialysis

1. Initial loading dose between 15 to 20mg/kg upon starting therapy
2. Vancomycin level in 48 to 72 hours (timing of level to align with their next HD session). Level to be taken **before** their dialysis on that day.
3. Depending on the vancomycin result, a vancomycin 'top up' dose may be given during the last hour of the dialysis session to account for vancomycin removal by high flux dialysis mechanisms
4. The 'top up' dose given will depend on the vancomycin level preceding the dialysis session, and the known removal of ~30% by high flux dialysis
5. As a general guide, the 'top up' dose will range from 500mg to 1000mg with consideration of the pre-dialysis vancomycin result

CRRT

Continuous Venovenous Haemodiafiltration (CVVHDF)

Degree of vancomycin removal will be influenced by the filter type and flow rates employed

In general

1. Consider an initial loading dose of 15 to 20mg/kg
2. Subsequent doses of between 1g to 1.5g q24h
3. Trough level to be measured before the 4th or 5th dose

PERITONEAL DIALYSIS

Patients receiving peritoneal dialysis

1. Initial standard loading dose of 2000mg upon starting vancomycin therapy
2. Levels should be repeated approximately every 72 hours to assist with determining when to re-dose
3. A subsequent dose to be given when the vancomycin level falls to $\leq 15\text{mg/L}$
4. In practice, the most common dose prescribed when redosing is 2000mg